DanceScape: Exploring Movement-to-music Interface for Children

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Music creation gives children a space to express themselves, yet opportunities to compose original music remain scarce. Although digital audio workstations (DAWs) offer powerful features, the complexity and steep learning curve of DAWs create significant barriers that prevent most children from using them. To address this gap, we developed DanceScape, an interactive movement-to-music interface that lets children control and shape music playback through embodied movement. Using a webcam, DanceScape recognizes users' movements, extracts tempo and dynamic cues, and adjusts musical output in real time. In a user study with 23 children at local exhibits, participants demonstrated strong engagement with and positive reactions to DanceScape. Participants with prior music-creation experience suggested using embodied gestures to control expressive parameters, such as brightness (filter cutoff) and effects (reverb, delay), rather than precision-based controls.

CCS Concepts: • **Human-centered computing** → **Interactive systems and tools**; **Human computer interaction (HCI)**.

Additional Key Words and Phrases: Children, Dance, Tangible & Embodied Interaction, Music Creation

1 Introduction

Music creation is emerging as an important part of contemporary music education as it provides several benefits [26]. For example, music creation experiences can contribute to children's socio-emotional development [5, 10, 15, 28, 36]. Also, the process of music creation naturally calls for agency, the ability to make choices [24] and act on them while having the initiative to work toward specific goals [35]. Despite these recognized benefits, prior studies consistently showed that existing school music programs do not provide children with enough opportunities for music making [14, 20, 40].

Children may have creative musical ideas but often have not yet mastered the technical skills in musical notation and instrumental performance required to translate these ideas into concrete musical forms. While Digital Audio Workstations (DAWs) are potential solutions for learners who do not identify as traditional musicians[33], children still struggle with a frustrating gap between musical imagination and technical execution [12]. Access barriers to DAWs remain a primary constraint for music-making in contemporary school settings [6, 11, 23].

Research shows that children often express musical ideas through movement before they can articulate them verbally or notationally [18], yet most DAWs ignore this crucial developmental pathway. Therefore, we designed and implemented DanceScape, an interactive system that synchronizes music tempo with a user's embodied movement dynamics. DanceScape enables users to control music tempo through embodied movement captured via webcam, where the system quantifies movement dynamics and proportionally adjusts playback speed in real-time: faster movements increase tempo while slower movements decrease it, creating an intuitive interactive experience between physical expression and sound.

We ran an exhibit deployment study at a community festival and school field trip with 23 children. Through this exhibit study, we focused on investigating the following research questions:

- **RQ1**: What forms and patterns of movement do children demonstrate when using their bodies as an input to control musical playback?

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- **RQ2**: What kinds of musical elements do children want to control through embodied musical interaction?

We found that participants had positive reactions to DanceScape with strong engagement patterns. Participants with prior music-creation experience suggested using embodied gestures to control expressive parameters, such as brightness (filter cutoff) and effects (reverb, delay), rather than precision-based controls.

2 Background and Related Work

Children naturally express musical ideas through movement before they can articulate them verbally or through notation [18]. Embodied music cognition theory supports this "natural" coupling between music and movement [38], where motion patterns serve as expressive features in musical sound that resemble biological motion. Music and movement share fundamental time-based characteristics, and children naturally synchronize their bodies to rhythm without formal training [31]. Through this process, they learn to connect what they hear with how they move and what they want to express [19, 22]. Yet most music creation software, including Digital Audio Workstations (DAWs), relies on technical interfaces rather than this crucial developmental pathway, presenting barriers that create frustrating gaps between musical imagination and realization [6, 11, 12, 23].

Several pioneering works have explored translating embodied movements into musical elements, establishing important conceptual foundations for movement-based music systems. The Nervous System [34] divided spaces into tiles with designated sounds, SICIB [27] used body movements for composition, and MiMU Gloves [16] enabled gesture-based control. More recent work includes Bruen et al.'s system translating craft-making gestures into sounds using dynamic time warping for movement detection [8]. Bodymouth demonstrated movement-based expressive interfaces [30]. These systems successfully demonstrated the feasibility of movement-to-music translation and revealed possibilities for generating expressive musical responses. However, they were designed for different contexts and audiences, often requiring specialized hardware or focusing on performance rather than education. Building on these valuable precedents, our work addresses a distinct need: systems specifically designed for children's developmental needs and educational contexts that operate without additional hardware while supporting creative exploration rather than predetermined outputs.

3 DanceScape: Interactive Embodied Control of Musical Tempo Dynamics

We designed and implemented *DanceScape*, an interactive system that synchronizes music tempo with a user's embodied movement dynamics. The system captures the user's embodied movements through a webcam and processes the video frames to quantify movement dynamics. Using the real-time movement data, DanceScape automatically adjusts the music's playback tempo; as the user's movement becomes more energetic or vigorous within a given time period, the music tempo increases proportionally. This *movement-to-tempo* mapping creates an intuitive and interactive experience: users directly influence the music's pace through their physical expression, with faster movements resulting in faster music playback, creating a dynamic feedback loop between embodied movement and sound.

3.1 Interface Design

Our system, *DanceScape*, consists of two interface components: a digital interface and a physical controller. Below, we describe the design and purpose of these two interfaces in detail.

3.1.1 Digital Interface. We designed the digital interface as shown in Figure 1. This interface provides six key features: [A] a music selection panel that allows users to choose and play tracks; [B] a real-time webcam feed that displays users' movements with yellow tracking points highlighting four key joints (left and right wrists and knees); [C] an optional video

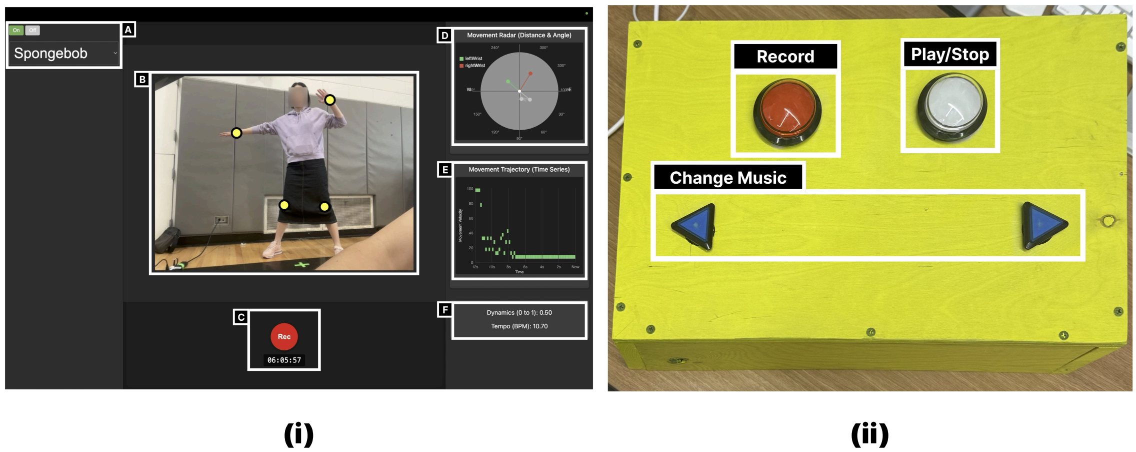

Fig. 1. The digital interface and controller of *DanceScape*. (i) Interface: users can observe their captured movements and visualized movement dynamics alongside the corresponding musical tempo changes in real-time, creating an immediate feedback loop between physical expression and audio output. (ii) Physical Controller: The physical controller of *DanceScape* provides four buttons to control the digital interface.

recording function; [D] a movement radar visualizing the relative position of each tracked joint; [E] a movement trajectory graph that displays the time-series velocity of embodied movements on a scale from 0 to 100 (where 0 represents minimal movement and 100 indicates highly dynamic movement); and [F] a real-time display showing current movement dynamics and the corresponding normalized music tempo. All metrics and visualizations are processed in real-time as users interact with the system.

3.1.2 Physical Controller. We designed a physical controller that allows users to control the digital interface without directly accessing the computing device (laptop). This controller serves two important purposes: first, it enables users to maintain their current physical position during interaction, eliminating the need to repeatedly approach the laptop for adjustments; second, it prevents accidental interactions with the system that might cause errors or disruptions.

Figure 1 shows the physical controller design, which features four control buttons. The *Record* button enables optional video capture of the user's performance; the *Play/Stop* button allows users to start or stop music playback; and the dual *Change Music* buttons (previous/next) provide navigation through the available music selections. We mounted these buttons on a durable plywood board, specifically designed for foot interaction during dance movements. To enhance visibility and encourage interaction, we colored the entire controller bright yellow. The controller is positioned on the floor within the dance area (see Figure 2).

3.2 System Implementation

We built DanceScape frontend using React [37] and integrated PoseNet [32], a computer vision model for real-time pose estimation, through the p5.js [25] library. To optimize performance and ensure responsive feedback in the web browser, we configured DanceScape to track four key body points: the left and right wrists and the left and right knees. This selective tracking approach reduces computational overhead while capturing the essential movement data required for musical expression and interaction.

We designed DanceScape to compute velocity changes using four key body points tracked across each frame. The algorithm first creates a velocity time series by measuring the distance traveled between frames divided by the time elapsed. The algorithm then applies a Fast Fourier Transform (FFT)[9] to extract the dominant frequencies from this time series. These frequencies determine the tempo, which is normalized to fall within an acceptable playback rate range (e.g., 0.25 to 2.0). Volume is calculated separately based on the standard deviation of the pose points at any given moment.

4 Study

4.1 Exhibit Descriptions

We exhibited a demo of the DanceScape system at a STEM+M (Science, Technology, Engineering, Mathematics, and Music) festival held at a local community center and school field trip in a suburban city in the United States. The festival ran from 10:00 am to 1:50 pm on a Sunday, providing a structured but flexible environment for families to explore a variety of fun and educational activities. The festival offered an opportunity to test our prototype with attendees who had the following characteristics:

- **Age**: Our study specifically targets young music learners (7-14 years old). This age range represents a critical period [21, 29] for music education when children have developed the cognitive and physical abilities necessary to engage with our interactive music activities.
- **Prior Music Creation Experience**: The event included children with varying levels of prior music creation experience, from complete beginners to those with experience with DAWs. Including children with previous music creation experience was especially valuable for addressing our research question about which musical parameters and features young music creators prefer to control through embodied movement.
- **Receptivity for Novel Music Creation Tools**: This event was centered on music and STEM activities. The participants were already able to think about the connections between technology and creative expression, so they could be receptive to exploring a new music creation tool.
- **Motivated Audience**: All of the activities at the festival were free-choice learning opportunities. This helped ensure that all participants came with motivation to explore—they were driven by personal interest rather than curriculum requirements.

4.2 Participants

A total of 23 participants engaged with DanceScape, of whom 19 completed the post-exhibit survey. Among these 19 respondents in grades 2-8, seven had prior experience with music creation. We provide details of the participants in Table 1. Among the 19 respondents, five reported extensive music creation experience using tools such as TunePad [13], GarageBand [4], and SoundTrap [1]. During our post-exhibit survey, we asked these five to identify which crucial DAW features (*e.g.,* Pitch, Duration, and Velocity) could be controlled through DanceScape. We had IRB approval to collect observational data, and all groups agreed to media recordings when entering the festival space.

4.3 Interaction Procedure

4.3.1 Instruction. We gave participants instructions for using the interface and the controller. The participants could choose a song to control with their dance moves. They could also see the timeline-based trajectory (dynamics and duration) and the movement radar: the 360-degree position of the tracking points of the body (left wrist, right wrist, left knee, and right knee). At the time of the festival, the controller buttons were unlabeled, so we explained how each button worked.

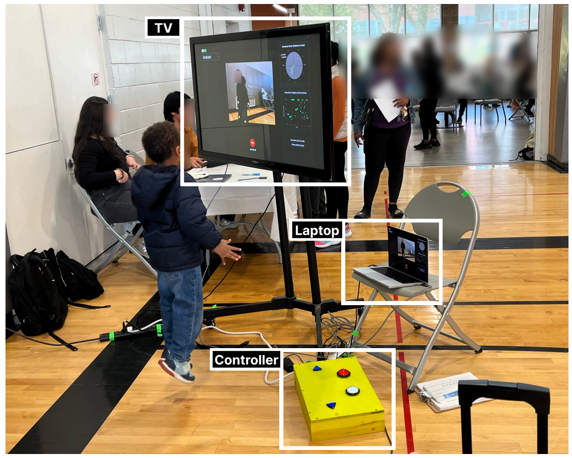

Fig. 2. Exhibit demo session design.

4.3.2 Interaction. We placed a marked starting point on the floor to ensure the camera could optimally capture participants' full-body movements. Participants began by selecting a sound to control and then positioning themselves at the designated starting point to begin interacting with the DanceScape system. They moved their arms and legs, spun around, laid on the floor, and jumped around, then invented other movements. We did not impose a time constraint on participants unless someone else was waiting for a turn. The presence of other attendees waiting to experience the system may have naturally influenced the duration of individual interactions. Otherwise, we allowed participants to engage with the system naturally until they felt ready to conclude their session. The average interaction time was 154.04 seconds ($\approx$ 2.57 minutes), with a standard deviation of 151.37 seconds ($\approx$ 2.52 minutes). This open-ended approach to session length was intentional, as it allowed us to observe authentic engagement patterns and gave participants sufficient time to explore the system's capabilities without artificial time constraints that might influence their experience or feedback.

4.4 Data Collection

Data were collected through two primary methods: video analysis with observational notes during the exhibit and post-exhibit surveys. Video was recorded while two researchers made simultaneous notes and observations with the consent of the participants and their guardians. This observational documentation allowed for a detailed analysis, as presented in Section 5.1. We also conducted a post-task survey with consenting participants to gather quantitative and qualitative feedback on the system. The survey included three key areas: (i) general satisfaction ratings to quantitatively measure user experience, (ii) specific aspects of the system that participants found appealing, and (iii) desired future features or modifications to inform system improvements.

4.5 Analysis

4.5.1 Video Analysis. We analyzed video recordings to characterize participants' interactions with DanceScape (RQ1). For the video analysis, we employed a systematic approach to examine participants' interactions with the tool. Each participant's session was divided into discrete temporal chunks, which two researchers analyzed collaboratively to ensure consistency and reduce individual bias. We focused on several key dimensions: interaction types (documenting which

Table 1. Participant Overview (N=23): Of the 23 participants who engaged with DanceScape, 19 completed the post-exhibit survey, with 7 reporting prior music creation experience.

#	Grade	Duration(sec)	Survey	Prior Music Exp.	#	Grade	Duration(sec)	Survey	Prior Music Exp.
P1	7	45	Y	N	P13	7	460	Y	N
P2	5	50	Y	Y	P14	6	155	Y	N
P3	N/A	118	Y	N	P15	8	394	Y	Y
P4	2	141	Y	N	P16	6	174	Y	Y
P5	5	86	Y	Y	P17	7	20	Y	N
P6	7	180	Y	Y	P18	7	150	Y	N
P7	5	113	Y	Y	P19	8	101	Y	N
P8	3	189	Y	N	P20	N/A	651	N	N/A
P9	2	70	Y	N	P21	N/A	143	N	N/A
P10	5	81	Y	N	P22	N/A	40	N	N/A
P11	5	72	Y	Y	P23	N/A	70	N	N/A
P12	7	40	Y	N					

body parts participants moved and how they engaged with the system), session duration and temporal patterns, musical selection preferences (categorizing the types of sounds or songs participants chose to control), pre- and post-activity questions or comments that revealed participants' expectations and reflections, and exploratory gestures that demonstrated how participants investigated the interface capabilities. This multi-dimensional coding approach allowed us to capture both the physical aspects of embodied interaction and the cognitive processes underlying participants' engagement with the DanceScape system.

4.5.2 Survey analysis. We analyzed interview and survey responses to learn participants' preferences and impressions on controling music playback through embodied movement (RQ2). For qualitative responses to the survey, we conducted a thematic analysis to categorize and analyze participants' open-ended feedback [7]. This approach involved identifying, analyzing, and reporting patterns within the data to understand participants' experiences and preferences regarding embodied music creation. For quantitative responses, including the star ratings and yes/no questions, we tabulated the data and calculated frequency counts for each response category. The thematic analysis process, including initial coding, theme development, and final categorization, was documented and visualized using Miro to ensure transparency and reproducibility in the analytical process.

5 Results and Discussion

Based on video recording analysis, we first characterize user engagement with DanceScape in exploratory interactions and social interactions (RQ1). Then we identify user preferences and impressions of controlling music through embodied movement based on participants' interviews and survey results (RQ2). Each participant spent an average of 154.04 seconds ($\approx$ 2.57 minutes), standard deviation = 151.37 seconds ($\approx$ 2.52 minutes) interacting with the system.

5.1 Exploratory Interactions with DanceScape

Several participants (*e.g.,* P14, P20, P21) demonstrated various exploratory embodied movements, including progressively larger movements, accelerating motions, and diverse movement types. These exploratory behaviors often escalated into deeper engagement. For example, P14 returned to try the system again and remained still, waiting for the music to slow down. When the tempo decreased, P14 initiated whole-body movements and spinning motions. P20 particularly enjoyed high tempos and continued moving until nearly out of breath. P21 put down a snack to fully engage with the system, employing whole-body movements that became progressively larger, indicating deep immersion in the experience. However, other participants demonstrated decreasing interest in DanceScape after initial exploratory movements. This disengagement resulted from intrinsic factors, such as feeling uncomfortable or unwilling to dance, and extrinsic factors, including system detection limitations that failed to recognize subtle movements below the established threshold.

5.1.1 Experimenting Playful System Manipulation. Four participants demonstrated playful system manipulation. For example, P15 outwitted the system by moving their hand and fingers very close to the camera. P15 also repeated the same movements (e.g., spins) that the participant believed the system would recognize. P14 showed interest in the tempo display and wanted to achieve both the slowest and fastest possible speeds. P21 positioned themselves in front of the camera laptop and began controlling their movements. P21 extended a hand toward the camera, recognizing that the system was detecting these actions. P21 continued exploring until finding the desired music, then became extremely engaged, moving their whole body for over 30 seconds. Some participants also showed interest in the movement radar displaying the distance and angle of their wrists and legs.

5.1.2 Experimenting Embodied Movement. The system's visual feedback through tracking points and responsive music speed changes fostered a sense of interactivity that encouraged spontaneous dancing. Participants (9 out of 19) reported that this real-time connectivity between their movements and music made them feel more motivated to dance and created a uniquely engaging experience that went beyond passive music listening.

> P4: *I like it [with giggles]. It's really silly when it's in slow motion.*

> P7: *I like how it [this system] speeds up & slows down with my body movements.*

The results demonstrate that children intuitively connect embodied movements with music. This supports the idea that responsive music systems are naturally accessible to young users [2, 3]. Real-time feedback motivated dancing and emotional engagement, confirming that interactive systems amplify user involvement in musical activities.

5.2 Social Interactions with Other Audiences

5.2.1 Observing Other Participants' Interactions. Participants frequently engaged in social interactions with friends, family members, and passersby. For example, three siblings (P6, P8, P21) demonstrated collaborative interactions, experimenting together and mimicking each other's movements to observe how different actions affected the musical tempo. This phenomenon aligns with the *honeypot effect* [39], which describes how the presence and activities of people interacting with a system can passively attract observers and passersby, encouraging them to approach and eventually participate.

5.2.2 Collaborating with Other Participants. Participants wanted more collaborative features in DanceScape, showing aspects to improve for future system design. For example, P7 and P16 wanted to dance with multiple people while interacting with the music together.

> P7: *I wish DanceScape could involve more people.*

Multi-user systems could enhance collective engagement by enabling group dancing experiences. However, this would require thoughtful design considerations around how to meaningfully integrate multiple participants' movements into the music creation context.

5.3 Preferred Musical Parameters to Control through Embodied Movement

Five participants (P2, P5, P6, P7, and P15) with extensive music creation experience using DAWs such as GarageBand, Soundtrap, and TunePad provided valuable insights into embodied control for music production. Notably, they recognized potential value in movement-based control across the 12 features we presented; no feature was unanimously rejected, indicating broad acceptance of embodied interaction control in music creation.

5.3.1 Less Suitable for Embodied Control: Precision-Based Features. Four participants (80%) marked velocity, panning, and modulation as less suitable for movement control. One possible interpretation of their reasoning could be related to precision requirements: these features (i.e., velocity, panning, and modulation) demand fine-grained adjustments that traditional DAW interfaces handle effectively through precise sliders and knobs. Participants might feel confident in their current manual control methods for these parameters, suggesting that embodied movement might introduce unnecessary complexity rather than enhancement for precision-dependent tasks.

5.3.2 Highly Suitable for Embodied Control: Expressive Features. In contrast, three participants (60%) endorsed embodied control for timbre-related features, including filter cutoff (sound brightness), resonance (frequency emphasis), and sound effects (reverb and delay). They also identified volume control as well-suited to movement interaction. These features share a common characteristic: they benefit from expressive and continuous manipulation that naturally maps to human gesture. It implies that embodied control could offer more intuitive and dynamic interaction than traditional interface elements.

6 Limitations and Future Directions

The DanceScape design presents several limitations. First, DanceScape's limited library of pre-recorded music pieces constrained participants' musical experiences, despite our careful curation of diverse and engaging content. Participants wished for broader musical variety. Second, the tracking feature of DanceScape tracked only four joints (left and right wrists and knees), which restricted the range and expressiveness of possible movements. This limitation became apparent when participants (*e.g.,* P14) wanted more freedom in their dance interactions, suggesting that full-body tracking would better support natural movement expression. While these constraints were appropriate for our proof-of-concept prototype, future iterations should incorporate expandable music libraries and comprehensive motion capture to fully realize the potential of embodied music interaction.

Furthermore, our study design has several limitations. First, the participant recruitment was geographically constrained to a high-income area in the United States. This could potentially create socioeconomic bias since children from affluent families might have greater exposure to arts and music education [17], which may have influenced their receptiveness to our system. Future studies should recruit participants from diverse socioeconomic backgrounds to ensure broader generalizability. Second, the exhibition format limited our ability to conduct in-depth interviews, preventing us from exploring the underlying reasons behind participants' survey responses through follow-up questions or clarifications. Future research should incorporate structured interview protocols alongside user testing to capture richer qualitative insights and better understand participants' experiences with embodied music interaction systems.

7 GenAI Usage Disclosure

Gemini and Overleaf's built-in grammar checking functions were used for proofreading. Cursor Co-pilot was used during the system development phase to generate code suggestions and implementation recommendations.

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